

01

HARDWARE

7 LESSONS

PHASE 1 • FOUNDATIONS

Hardware, Computing & IoT Fundamentals

Understand the machines you are protecting. From transistors and the boot process to CPUs, memory, storage and the Internet of Things.

01

How Computers Really Work

BEGINNER

CONCEPT

The journey of a computation: from your keystroke to electrons moving...

20 min →

02

CPU, Memory & the Boot Process

BEGINNER

CONCEPT

Registers, cache, RAM, and the boot sequence – the essential foundati...

30 min →

03

Operating Systems in Depth

BEGINNER

CONCEPT

Kernel, user mode, processes, virtual memory, and system calls – the ...

30 min →

04

IoT & Embedded Systems Security

BEGINNER

LAB

Smart devices, microcontrollers, firmware and the special security ni...

35 min →

05

Virtualization, Hypervisors & Containers

BEGINNER

CONCEPT

VMs, hypervisors, and containers – the isolation mechanisms underpinn...

25 min →

06

Storage, File Systems & Forensic Basics

BEGINNER

LAB

How data is stored, what happens when you delete a file, and the fore...

30 min →

07

Network Hardware & Devices

BEGINNER

LAB

30 min →

Routers, switches, firewalls, APs, and the topology of the network – ...

[Networking & Protocols →](#)